

FILTER DESIGN TEST CASE

BAW (Bulk Acoustic Wave) - Band 2 (1900 MHz) for Satellite L-Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a BAW (Bulk Acoustic Wave) filter targeting Band 2 (1900 MHz) for Satellite L-Band Filter. The filter is designed on 128° YX LiNbO3 substrate with a target center frequency of 5082.1 MHz and bandwidth of 302.8 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Satellite L-Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: BAW (Bulk Acoustic Wave)
- Frequency Band: Band 2 (1900 MHz)
- Application: Satellite L-Band Filter
- Substrate: 128° YX LiNbO3
- Package: QFN (Quad Flat No-Lead)
- Design Tool: Synopsys CustomSim
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the BAW (Bulk Acoustic Wave) design targeting Band 2 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5082.1 MHz
Bandwidth (3 dB):	302.8 MHz
Insertion Loss Target:	<= 2.3 dB
Return Loss Target:	>= 22.0 dB
Out-of-Band Rejection:	>= 50.0 dB
Isolation (Duplexer):	>= 21.0 dB
Q Factor:	2718
Temperature Coefficient:	-22.2 ppm/°C
Die Size:	2.1 x 1.2 mm
Wafer Size:	6-inch
Number of Resonators:	11
Electrode Material:	W

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Synopsys CustomSim and load the BAW (Bulk Acoustic Wave) template project.
2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
3. Set design targets: center freq 5082.1 MHz, BW 302.8 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (QFN (Quad Flat No-Lead)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 128° YX LiNbO3).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 15246 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.3 dB
- Return loss in passband shall be >= 22.0 dB
- Out-of-band rejection at +/- 454 MHz offset >= 50.0 dB
- Group delay variation across passband shall be <= 6 ns
- Temperature drift over -40°C to +85°C shall be <= 14.1 MHz
- Power handling shall be >= +31 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 87%

6. TEST RESULTS

Measurement Date: 2024-11-11
Devices Tested: 68
Insertion Loss (meas): 2.02 dB
Return Loss (meas): 24.1 dB
Rejection (meas): 52.7 dB
Isolation (meas): 26.3 dB
Q Factor (meas): 2718
Group Delay Variation: 4.1 ns
Power Handling: +35 dBm
Die Yield: 81.0%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The BAW (Bulk Acoustic Wave) design demonstrated below-target performance for Band 2 (1900 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Group delay flatness improved compared to previous revision.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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